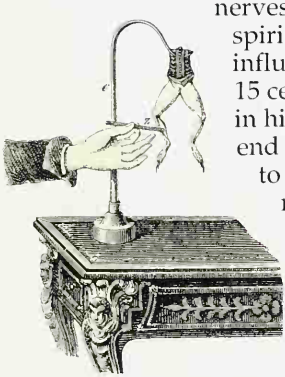
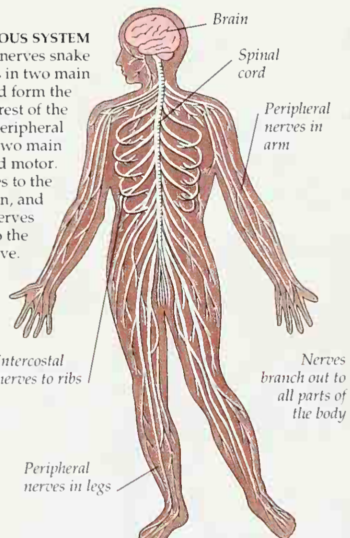
The nervous system

A WARNING SIGN
Microscopic and treelike, nerve endings are found in the skin and in many internal organs. They register discomfort and pain, from a broken bone to an aching tooth – or a pulled tooth. Pain warns that something is amiss, and that action should be taken to care for the body.

WITHOUT ITS TELEPHONE SYSTEM of cables snaking across the land and spreading through towns and cities, a modern country would soon be paralyzed. Without its nervous system, so would the body. Nerves are the communication network of the body. They carry messages to and fro, inform, and coordinate. Like telephone signals, nerve signals are electrical, though they are only about one-twentieth of a volt in strength. Herophilus of Alexandria was the first to notice that nerves are concerned with sensations and movements. Around 300 BC, Roman anatomist Galen recognized the nerves' roles in control and coordination, but he thought that nerves are hollow, and that some type of mystical "animal spirit" fluid passes through them. His mistaken theories influenced science and held back progress for a further 15 centuries. Vesalius mapped the nervous system, but not in his usual detail. However, he did show that nipping the end of an exposed nerve would cause the muscle attached to jump. With the discovery of the electrical nature of nerve signals in the late 1700s, the workings of the nervous system were at last becoming clearer.

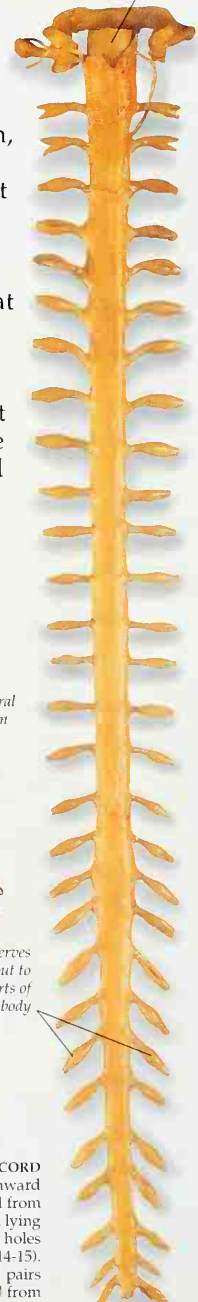


"ANIMAL ELECTRICITY"
Anatomist Luigi Galvani (1737-1798) of Bologna noticed that an isolated frog's leg twitched when metallic contact was made between its nerve and its muscle. He believed the nerves contained "animal electricity." Physicist and rival Alessandro Volta (1745-1827) showed the electrical nature of the nerve signal.



THE NERVOUS SYSTEM
Some 30,000 miles (50,000 km) of nerves snake through the body. The system is in two main parts. The brain and spinal cord form the central nervous system, and the rest of the body-wide network is called the peripheral nervous system. The latter has two main types of nerves – sensory and motor. Sensory nerves carry messages to the brain from the eyes, ears, skin, and other sense organs. Motor nerves convey signals from the brain to the muscles, making the body move.

PAVLOV'S PERFORMING DOGS
A reflex is an automatic reaction carried out by nerve signals. A well-known example is how dogs naturally salivate, or drool, at the sight and smell of a meal. Conducting experiments on the digestive system, scientist Ivan Pavlov of Russia (1849-1936) noted that dogs salivated at the sight of their handler, even without food. He trained them to associate food with the sound of a bell. After a time they drooled upon hearing the bell alone. This type of "re-programmed" auto-action is known as a conditioned reflex.



THE SPINAL CORD
The spinal cord is, in effect, a downward extension of the brain. It is well protected from damage by knocks, kinks, and pressure, lying inside the long tunnel formed by the holes through the vertebrae, or backbones (pp. 14-15). From its 16-in (40-cm) length branch 31 pairs of nerves, each carrying signals to and from a certain part of both sides of the body.